Linear Algebra and Calculus Lecture 11

Eric Sandin Vidal

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Null Space and Column Space

In this lecture we will look at certain important vector subspaces.

Definition. The **null space** of an $m \times n$ matrix A, written as $\operatorname{Nul} A$ is the set of all solutions of the homogeneous equation ______. In set notation,

$$\operatorname{Nul} A = \{x \,:\, x \in \mathbb{R}^n, \underline{\hspace{2cm}}\}.$$

In other words, the null space of A is the set of points that get mapped to the 0 via the linear map $x \to$ ___.

Theorem. *The null space of an $m \times n$ matrix A is* _______________.

Exercise. Prove it!

Example. Let's find a spanning set for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \tag{1}$$

The first step is to find the general solution of $A\mathbf{x} = 0$ in terms of the free variables. We row-reduce the augmented matrix $[A \mid 0]$ to reduced echelon form:

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

From the row-reduced matrix, we get the following system of equations:

$$x_1 - 2x_2 - x_4 + 3x_5 = 0$$

$$x_3 + 2x_4 - 2x_5 = 0$$

The general solution is:

$$x_1 = 2x_2 + x_4 - 3x_5, \quad x_3 = -2x_4 + 2x_5$$

with x_2, x_4, and x_5 free. Now, decompose the general solution vector into a linear combination of vectors where the weights are the free variables:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

Thus, the spanning set for Nul A is:

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Definition. The **column space** of an $m \times n$ matrix A, written as Col A, is the set of all linear combinations of ________________. If $A = [a_1 \cdots a_n]$, then

$$\operatorname{Col} A = \operatorname{Span}\{a_1, \ldots, a_n\}.$$

Theorem. *The column space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^m$.*

Proposition. *The column space of an $m \times n$ matrix A is all of $\mathbb{R}^m$ if and only if ____________ ______________________________________.*

Example. Let's find the column space of the matrix in (1). We already had it row reduced:

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Since x_2, x_4 and x_5 are free, the pivot columns are the first and third columns, therefore the column space of A is spanned by

$$\left\{ \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix} \right\}.$$

Notice how we took the columns in the original matrix, not in the row reduced one!

Remark. These processes for finding null spaces and column spaces produce a linearly independent spanning set.

Remark. Notice how simple it is to verify if a vector is in Nul A and how more laborious it is to find a spanning set for it, and how laborious it is to verify if a vector is in Col A but how simpler it is to find its spanning set.

Linear Transformations

Definition. Let V and W be vector spaces. A **linear transformation** $T : V \to W$ is a map such that

1. $T(u + v) =$ __________, for all $u, v \in V$,
2. $T(cu) =$ _____, for all $u \in V$ and all scalars $c \in \mathbb{R}$.

Definition. The **kernel** of a linear map T ($\ker T$) is the set of all $u \in V$ such that ________.

Definition. The **range** of a linear map T ($R(T)$) is the set of all vectors in W of the form ____ for some $x \in V$.

Proposition. *If a linear map T is given by a matrix transformation with standard matrix A, then* $\ker T =$ _____ *and* $R(T) =$ _____.

Linear Independency

The idea of linear independence of general vector spaces is the same as for $\mathbb{R}^n$. An indexed set of vectors $\{v_1, \dots, v_p\}$ in V is said to be **linearly independent** if the vector equation

$$\underline{\hspace{6cm}} = 0$$

has only the trivial solution _______________. If some of the weights are non-zero, then the set is **linearly dependent**.

Theorem. *An indexed set $\{v_1, \dots, v_p\}$ with $p \geq 2$, with $v_1 \neq 0$, is linearly dependent if and only if some v_j (with $j > 1$) is a linear combination of the preceding vectors $v_1, \dots, v_{j-1}$.*

Bases

Definition. Let H be a subspace of a vector space V. A set of vectors $\mathcal{B} \subset V$ is a **basis** for H if

1. $\mathcal{B}$ is _______________________, and
2. the subspace spanned by $\mathcal{B}$ coincides with H, that is, ___________.

Example. Let $S = \{1, t, t^2, \dots, t^n\}$. The set S is called the standard basis for $\mathbb{P}_n$, since it spans the space and is linearly independent.

Example. Let

$$v_1 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 6 \\ 16 \\ -5 \end{bmatrix}, \quad \text{and} \quad H = \operatorname{Span}\{v_1, v_2, v_3\}$$

Note that $v_3 = 5v_1 + 3v_2$, and let's show that $\operatorname{Span}\{v_1, v_2, v_3\} = \operatorname{Span}\{v_1, v_2\}$. We will also find a basis for H.

Every vector in Span$\{v_1, v_2\}$ belongs to H because

$$c_1v_1 + c_2v_2 = c_1v_1 + c_2v_2 + 0v_3.$$

Now, let x be any vector in H — say,

$$x = c_1v_1 + c_2v_2 + c_3v_3.$$

Since $v_3 = 5v_1 + 3v_2$, we may substitute to get:

$$x = c_1v_1 + c_2v_2 + c_3(5v_1 + 3v_2)$$

$$= (c_1 + 5c_3)v_1 + (c_2 + 3c_3)v_2$$

Thus, x is in Span$\{v_1, v_2\}$, so every vector in H already belongs to Span$\{v_1, v_2\}$. We conclude that H and Span$\{v_1, v_2\}$ are actually the same set of vectors. It follows that $\{v_1, v_2\}$ is a basis of H, since $\{v_1, v_2\}$ is obviously linearly independent.

Theorem. *(The Spanning Set Theorem) Let $S = \{v_1, \ldots, v_p\}$ be a set in a vector space V, and let $H = \text{Span}\{v_1, \ldots, v_p\}$.*

1. *If one of the vectors in S — say, v_k — is a linear combination of the remaining vectors in S, then the set formed from S by removing v_k still spans H.*

2. *If $H \neq \{0\}$, some subset of S is a basis for H.*

Remark. The processes shown before for finding spanning sets for null spaces produce automatically a basis for the null space, since the set is linearly independent.

Similarly for the column space:

Theorem. *The pivot columns of a matrix A form a basis for _____.*

When the Spanning Set Theorem is used, the deletion of vectors from a spanning set must stop when the set becomes linearly independent. If an additional vector is deleted, it will not be a linear combination of the remaining vectors, and hence the smaller set will no longer span V. Thus a basis is a spanning set that is as small as possible. A basis is also a linearly independent set that is as large as possible. If S is a basis for V, and if S is enlarged by one vector—say, w—from V , then the new set cannot be linearly independent, because S spans V, and w is therefore a linear combination of the elements in S.